

The Engine at the Waist

Nouns above, roads below. What in the parked render engine has to exist as code, what should be worn as facets in ECS land — and the old-world instance sitting on each class that still holds.

You click a card. Today: `selection/transition` (a case on `:toggle`) → `chrome_derive/apply-selection` → five form-nodes → `chrome_material/material-quads` (a case on `:selection-outline`) → `chrome_gpu` vertices → tape entries carrying `:pick {chrome-form ...}`. Six files know the words *selection-outline* and *blue*.

Above the waist: the pointer system puts `:selected` on the card entity; a recipe — data — says *selected things wear a 1 px blue outline and four 10 px corner marks*; the floor draws any `:mark` it is handed. No file knows the word `selection`. The blue is yours to change in-land, and a caret, a hover halo, a focus ring are the same recipe with different numbers.

ABOVE THE WAIST — the old page's instances · every box here goes · reborn above as recipes (data) + systems (code over facets)

git keeps the geometry

THE TREE — "everything is a rect"

goes · your ruling

rect_tree.cljc · scene_store.cljc
scene_runtime.cljs · selection.cljc

one node, a :data slot per kind, eight walkers;
pick by bounds; store slots by vi; "rect = blocks".

reborn as: entities wearing :bounds; layout = a system
(80 lines, shape-blind, re-derived over facets).

MARKS VOCABULARY — the code says "chrome"

goes

chrome_material forms · chrome_derive.cljc
snap.cljc · chrome_runtime.cljs

five forms + px constants; selection + gesture → marks;
two providers set by mutation; runtime dead without a host.

reborn as: a recipe — "selected wears a 1 px outline
and four 10 px corners"; the blue is a number in data.

ARROWS — connector family

goes · position, vetoable

connector_material.cljc · connector_route.cljc
connector_gpu.cljs

elbow Z-route, heads, boundary clip, a live pick mirror;
clones a text system for its label — the facet receipt.

reborn as: a route system that emits a :contour facet;
the pure geometry is listed by anchor for lifting.

THE SMALL BLENDER — the 3D editor

goes · either way

region3d_runtime.cljs · region3d_pointer.cljc
_scene: orbit · gizmo · handle picks
_material: kind menus · 9 edit ops
_gpu: gizmo + grid shaders · !pick-state

a draggable panel with orbit, gizmo, lights menu, grid.

reborn as: recipes + systems over :mesh / :transform;
the kernel under it (below) waits on your one question.

DEMOS · ORPHANS · AMBIENT STATE

goes

frame_runtime.cljs (8 demo trees, PNG chord)
region3d_fixture · ?live-atoms ?region3d readers
public/faces/ · gpu-mount ("DO NOT WIRE")
maintain-frame-plan · maintain-effect-spans
!effective-provider · !live-camera-provider
__softlandFrameLedger (js/globalThis)
window.region3d · window.sceneContext

the old page's eyes; maintained twins that lost to
the "view namespaces; providers set by mutation;
browser globals. (Oracles are not orphans — they stay.)

EDITOR LEFTOVERS inside renderer.cljs

goes

calculate-bracket-rects · hit-test
make-snapper · draw-comparison-frame!
create-editor-state (50 000 rects)

the old editor's hands left inside the GPU file:
bracket highlight, mouse → line/col, grid snapping.

renderer.cljs itself stays — it is the conductor,
below.

THE WAIST

render packet: road id + resource key ✓

family key → facet key · build stage

pick walks the tape backward ↑

1 recipes + facets → packets ↓

:paint/source carries a live handle ✗

placement reads the old slot ✗ · chrome_gpu wants a form-free input ✗

2

BELOW THE WAIST — the render floor · what has to exist as code · never editor code

pure .cljc runs on the JVM · .cljs owns the GPU

THE KEY — today: ten families rect · shadow · msdf · slug · clip · image · path · connector · chrome · region-3d · wanted: facets, several per entity

lives in: scene_tape registrations · frame_inputs table · renderer registry + buckets · scene_store · rect_tree walkers · selection / chrome_derive

build stage — not a cut

SHAPERS — said → geometry · pure · JVM-tested · the yardstick is text

TEXT — the yardstick

floor

text_shaper.cljs
text_layout.cljc
text_layout_planes.cljc
runtime/fonts.cljs

string → glyph runs (font units)
→ lines + positions, material-local;
the cache is a value; caret / hit
readers stay — nobody else measures.

open: legacy grid road (a JVM test).

STROKE / SHAPE

floor

path_material.cljc
path_tessellation.cljc

contours, or centerline + pressure,
→ triangles. No editor code at all.
You named this road yourself:
"stroke shape → tessellator →
triangles".

small: point-edit verbs in the
description ns (move-knot ...).

IMAGE · TRANSFORM

floor

image_material.cljc
containers.cljc

digest + size → atlas quad; the
registry and atlas are values in,
values out; no per-node transform.
spec → one affine, composed through
parents; forward / inverse point;
feeds every pipeline.

3D KERNEL

open — yours

region3d_scene.cljc
region3d_placement.cljc
region_rungs.cljc

primitive → mesh; parent table →
matrices; triangles → BVH; ray →
object; text / ink onto a plane.

stays iff an entity may wear :mesh.

✗ placement reads the old slot shape.

EFFECTS · ORDER

floor

scene_tape.cljc
frame_effects.cljc
frame_plan_view · frame_delta
frame_semantic_state.cljc
frame_graph.cljc = the oracle

decls + order → spans → passes;
the tape: one ordered list — paint
walks forward, pick backward,
"no family branch".

PIPELINES — geometry → pixels · nine, as-is · they do not change

GLYPHS — msdf · slug

inside renderer.cljs

two backends, one facet;
chosen downstream by the font.

CONTOUR

path_gpu.cljs

pure prepared-op → pack-vertices;
one case only — the clipper.

BOX · SHADOW · IMAGE

inside renderer.cljs

formula roads (a rounded box is
a per-pixel distance) + the quad.

3D PASSES

region3d_gpu.cljs
region3d_placement_gpu.cljs
depth · shading · placed glyphs;
→ a compositor lease. Grid +
gizmo overlays go (above).

MARKS — hybrid metric

chrome_gpu.cljs

anchor in world + offset in px —
a mark that never scales with zoom.
✗ takes the five forms today.

COMPOSITOR + CLIP

compositor_gpu.cljs

passes · targets · blur / mask /
opacity · scissor.
→ pixels

CUSTODY — the machine that owns GPU memory and order

THE CONDUCTOR

floor

renderer.cljs (203 KB)

frame compile / execute · the registry +
fence · shape-text (a real shaper) ·
shaders ≈ 18 % of the file.

the editor leftovers go (above); the
registry's key changes at build stage.

POOL · BUDGET · LEASES

floor

buffer_pool.cljs · gpu_budget.cljs
region_bindings.cljs

slots + generations · bytes accounted ·
region leases with epochs. Device-owned
state that must exist somewhere — stays.

gpu-mount ("DO NOT WIRE") goes.

DECLARATIONS

changes with the key

frame_inputs.cljc

per-family input table + entry deltas;
the quantized camera rungs.

✗ the ledger written to js/globalThis
goes.

STATICS — sorted, not lumped

stay:

device WeakMaps · buffers + capacity ·
bounded caches · leases · frame history

go:

providers set by mutation · browser
globals · the runtimes' session atoms

THE GUARD — receipts for the pipelines

verifier.cljs (304 KB) · test/render_engine/run_verifier.mjs · gpu-goldens/ (67 files) — its own shadow-cljs module; the only thing that boots the engine today

CPU oracles stay — reference geometry for the pipelines (region3d-lit-oracle · point-in-curves? · route-boundary-distance) — with the pipeline goldens.

Fixture atoms of a cut family go with it (an estimated 60–70 % of the file is scene construction — sampled, not counted); the W4 receipt is rehomed.

draw-frame! has no caller in src; the verifier reaches the GPU by composing compile-frame-tape + execute-frame-entry! itself.

□ floor — stays as code [---] goes — cut now, reborn above □ open — yours, or build stage ● 1 packets down · 2 pick back up
KB are file sizes; splits inside files are estimates; nothing here is a ruling

The engine at one grain. Above the line: the old page's instances, every box goes and is reborn above as data or as a system over facets. On the line: what crosses it today and what must not. Below: the render floor in three rows — shapers, pipelines, custody — and the guard.

The story

1. The engine is 46 files, about 1.48 MB, and it is **dark**: the only thing that boots it is the verifier's own shadow-cljs module `shadow-cljs.edn:15-16`; `draw-frame!` has no caller in `src/ renderer.cljs:3678`. Every call on this page is on *shape*, not liveness — your criterion.
2. Your test for the waist (08-19): *"something that has to exist as code vs what can and should be built on top, ecs or whatever style."* Smallest version: *"text in placement out."*
3. One refinement both reviews forced, and it matches your three strata (LOG 07-30: material · Softland code · host floor): **"has to exist as code" is not the same as "below the waist."** ECS systems are code too — a route solver, a layout rule, selection → marks can all be code *above* the waist. The floor is narrower: what is reusable without knowing Softland's meanings or gestures, takes and gives stable geometric values, or owns GPU resources.
4. The picture above is the engine at one grain. Below the line, three rows: **shapers** (said → geometry — pure `.cljc`, runs on the JVM, has tests), **pipelines** (geometry → pixels — `.cljs`, stateful), **custody** (the machine that owns GPU memory and order). Above the line: the old page's instances — every box marked *goes*.
5. Read the shapers row: it exists today, family by family — text (your yardstick), stroke, image + transform, the 3D kernel, effects + order. Not all of it is floor (see 3); the table below sorts it.
6. Read the pipelines row: **nine, as-is** — glyphs (msdf, slug), contour, box / shadow / image, the 3D passes, marks (the code says chrome), the compositor with clip. They do not change in anything below.
7. Here is the old world: the engine is **keyed by family** — ten nouns, a closed menu, the same ten words at every layer. A family fuses *a kind of thing* with *a pipeline* and *a grammar* (pick modalities, edit operations, hit tolerance, zoom regimes). An entry belongs to exactly one.
8. The fence that throws at load `renderer.cljs:3364` is *not* the class system — it is a completeness law any registry needs (every facet an entity may wear must have a road).

What closes the world is the **key**. Keep the fence, change the key.

9. The receipt that the key is wrong: a labelled arrow is a connector family that **clones a text system** `connector_gpu.cljs:185-226`, because one entry can be one family only. In your ECS words an entity wears several facets at once — `:text` and `:contour` and `:mark` — and each road takes its own.
10. The engine half-knows this: the tape's `paint-forward` "contains no family branch" `scene_tape.cljc:749`; the renderer's per-family path is a data table, not a `cond` `renderer.cljs:3297`. What is left of the class system is the key, the grammar per family, and the frame description bucketed by family.
11. The instances sitting on it: the box tree (you ruled: goes), the five mark forms + snapping + marquee + selection, the elbow-arrow conventions, the small Blender inside the 3D window, the demos — and, cross-cutting, ambient providers set by mutation, browser globals, and kind menus guarded by refusing validators.
12. The move, at its real size: **connector stops being a family** and becomes a system above; **the registry key goes from family to facet; an entity can wear several facets**. The pipelines stay. The named roads (`:text` `:contour` `:formula` `:image` `:mesh` `:mark`) are a target vocabulary, not a source fact — each needs a small contract before the registry changes.
13. So: what we need from the engine = the floor shapers, the pipelines, the order / passes / upload / budget custody, the guard's oracles. What we don't = every named kind and the key that names them. What waits = the key change — until the instances are gone and the renderer can be seen at its own grain.

The waist, drawn

ABOVE — THE OPEN WORLD

entities · facets · recipes · ECS systems · what a gesture means · arbitrary new combinations

Material is data. Systems are code — keyed by the facet they serve, never by what a thing *is*.

↓ value-only render packets — a road id + a resource key, never a live handle ↓

BELOW — THE FINITE MACHINE

road registry (exhaustive, fenced) · geometry and coverage preparation · GPU resource ownership · order, passes, composition · buffers, upload, execution

Reusable without Softland's meanings; stable geometric values in and out; the GPU is never handed invalid data.

Three strata, in your words (LOG 07-30, "the code editor is Softland at zoom 100"): **material** — data · **Softland code** — the systems, code above the waist · **host floor** — what must remain outside the land. In this phase the systems stratum has no home of its own yet, so the cut is still binary: floor code stays in the repo as engine; family-shaped code goes; git keeps whatever pure geometry is worth lifting later.

What the old world is here

TODAY — keyed by family (10 nouns; each = kind + pipeline + grammar)

:rect	→ rounded-box pipeline	
:shadow	→ shadow pipeline	
:msdf	→ glyph pipeline (atlas)	┌ one facet,
:slug	→ glyph pipeline (curves)	└ two backends

WANTED — keyed by facet

entity wears several facets;
each road takes one:

:text	→ glyph road
:contour	→ triangle road

```
:image      → image pipeline
:path       → contour pipeline
:connector  → (a route SYSTEM, above) → contour pipeline
:chrome     → marks pipeline
:region-3d  → 3D passes → compositor lease
:clip       → scissor
one entry = one family · grammar per family · fenced
```

```
:formula    → formula road
:image      → image road
:mesh       → mesh road
:mark       → mark road
:effects    → compositor
:order      → the tape
recipes (data): card · arrow · outline · 3D window
```

Where the key lives — the places a facet key would replace it: `scene_tape.cljc:14` (`family-ids`) and `:480` (ten registrations carrying pick modalities and edit operations) · `frame_inputs.cljc:28-69` (the declaration table) · `renderer.cljs:3297` (the executor registry) and `:3418` (the frame description's buckets: `:rects` `:shadows` `:images` `:paths` `:connectors` `:chromes` `:regions`) · `scene_store.cljc:319` (`derive-store-frame` fills the buckets) · `rect_tree.cljc:205-611` (eight walkers, one per kind) · `selection.cljc:12,145` and `chrome_derive.cljc:17-20` (family sets deciding behaviour) · the verifier's per-family atoms.

The one waist violation inside the tape itself: an entry's `:paint/source` carries a **live JS system object** `path_gpu.cljs:367-368` · `region3d_gpu.cljs:1401`. Data holding a mutable handle — an entry cannot cross the waist as a value today. The future packet names a road and a resource key; runtime-owned state is resolved below.

Every family, three bands

KB are file sizes (fact). Splits inside a file are estimates. Pills: `FLOOR` stays as code below · `SYSTEM` ↑ / `RECIPE` ↑ belongs above (code or data) · `GOES` cut now · `OPEN` not yet closable.

FAMILY	SAID → GEOMETRY (THE SHAPER)	PIPELINE	THE OLD-WORLD INSTANCE ON IT	WHERE IT SORTS	KB
text	<code>text_shaper</code> — one shaper (HarfBuzz wasm :13): string + features → glyph runs in font units :268-290. <code>text_layout</code> — + size, width, wrap → lines, glyph positions, metrics in material-local space :682-1000; the cache is a value the caller threads :1029-1084. <code>planes</code> — typed-array storage :290-304. The caret / selection / hit <i>readers</i> :1304, :1513 stay with it — nobody else measures.	msdf or slug, chosen downstream <code>fonts.cljs</code>:203-207	A second layout road, <code>legacy-layout</code> (monospace grid, 138 lines :174-311) — but at least one JVM test asserts it <code>text_layout_test.clj</code>:151: it is the JVM's stand-in shaper. A closed 16-keyword <code>ground-op-roles</code> :23-27.	FLOOR OPEN legacy road	125
box	<code>rect_tree/layout-children</code> — children + direction / gap / padding / align → bounds; a pure pre-pass :38, :54-132, shape-blind, ~80 lines.	rounded box + shadow, inside the renderer	The tree: node <code>{:id :type :bounds :style :children :data ...}</code> <code>rect_tree.cljc</code>:19-32 with one <code>:data</code> slot per kind and eight walkers; store slots by <code>vi</code> <code>scene_store.cljc</code>:96-126; pick by bounds :404; <code>selection</code> bound to <code>vi</code> /address with "rect = blocks" <code>selection.cljc</code>:145.	GOES the tree (your ruling) · SYSTEM ↑ layout — goes with the tree now; 80 lines to re-derive over facets	79
stroke / shape	<code>path_material</code> — contours, or centerline + pressure :8-10, :81 → <code>path_tessellation/tessellate</code> [<code>material zoom</code>] → triangles :383-402. Pure. No editor code. You named this road yourself.	contour — <code>path_gpu</code>, pure <code>prepared-op</code> → <code>pack-vertices</code> :217-252; its one <code>case</code> is a clipping algorithm :154	Point-edit verbs inside the description ns (<code>move-knot</code>, <code>move-contour-point</code> :288, :336) — small. The live handle in <code>:paint/source</code> <code>path_gpu.cljs</code>:367.	FLOOR	63
image	<code>image_material</code> — digest + size → atlas placement + a 13-word quad :257, :325; registry and atlas are values in, values out :67, :250.	image, inside the renderer	Almost none — the packer refuses per-node transforms by design :326-327.	FLOOR	15

FAMILY	SAID → GEOMETRY (THE SHAPER)	PIPELINE	THE OLD-WORLD INSTANCE ON IT	WHERE IT SORTS	KB
transform	<code>containers</code> — spec → one affine composed through parents; forward / inverse point :87, :232-253. Pure; highest fan-in in the tree.	feeds every pipeline	none	<code>FLOOR</code>	12
marks (code: chrome)	<code>chrome_material/material-quads</code> — world anchor + px offset → quads :241-257 — but it switches over the five forms, and <code>chrome_gpu</code> calls it <code>chrome_gpu.cljs:169</code> .	the hybrid-metric vertex: anchor in world, offset in px <code>chrome_gpu.cljs:182-204</code>	The five forms <code>#{:selection-outline :handle :marquee :guide-line :gap-tick}</code> <code>chrome_material.cljc:16</code> ; px constants (snap 8, handle 10, slop 6 :10-14); <code>snap</code> (guides, gap ticks); <code>chrome_derive</code> (selection + gesture → marks; two defonce providers :23-24); <code>chrome_runtime</code> (dead; throws without host hooks :330-333).	<code>FLOOR</code> the hybrid-metric pipeline · <code>OPEN</code> a form-free mark road is new work · <code>GOES</code> the rest, now	75
arrows	No shaper — <code>connector_route</code> is a system : bound thing → its boundary quad → clip from center :173 → elbow Z-route :185-195 → trim for heads :207, :218 → label at <i>t</i> :390 → synthesize a path material :334 → tessellate :360.	contour (borrowed) + its own buffer + a cloned text system <code>connector_gpu.cljs:185-226</code>	The whole family: <code>:elbow/v1</code> , triangle heads <code>connector_material.cljc:14-17</code> , a live pick mirror + provider by mutation <code>connector_route.cljc:18-21</code> , a 142-line per-edge cache :544-685, twelve system atoms <code>connector_gpu.cljs:115-126</code> .	<code>SYSTEM ↑</code> · position : cut the family whole now; the pure geometry above is listed for lifting from git when arrows are rebuilt over facets. Codex holds this open; one line from you vetoes it.	74

FAMILY	SAID → GEOMETRY (THE SHAPER)	PIPELINE	THE OLD-WORLD INSTANCE ON IT	WHERE IT SORTS	KB
3D	<code>region3d_scene</code> — primitive → mesh :325, parent table → matrices :140-160, triangles → BVH :394, ray → object :516, :947. <code>region3d_placement</code> — 2D text / ink onto a plane, reusing the text shaper :180, :334 — but it reads the old slot shape :22, :41, :60, :72, :78. <code>region_rungs</code> — texture budget grant :50-93.	region passes (depth, shading, placement glyphs / ink, grid, gizmo overlays) → compositor lease <code>region3d_gpu.cljs</code>:1122-1387	The small Blender: kind menus <code>region3d_material.cljc</code>:18-22, nine edit ops by <code>:op/id</code> :82-86, :511, orbit / gizmo / handle picks <code>region3d_scene.cljc</code>:616, :696, :932, gizmo + grid shaders <code>region3d_gpu.cljs</code>:257-378, :206, a global <code>!pick-state</code> + device WeakMap :705-706, <code>region3d_runtime</code> (panel at [236 146] , fixture scene, window.<code>region3d</code> :23, :78-118, :602-620), <code>region3d_pointer</code> (dolly / wheel), a CPU PBR reference only the verifier calls <code>region3d_scene.cljc</code>:1141 · <code>verifier.cljs</code>:4587.	OPEN the kernel (mesh · hierarchy · BVH · ray · placement) — your question · GOES the small Blender, either way · OPEN placement's input after the store goes	~260
effects + order	<code>frame_effects/derive-effect-spans</code> :204-227; <code>scene_tape/compile-tape</code> :724 + <code>paint-forward</code> / <code>pick-reverse</code> :748, :757; <code>frame_plan_view</code> (the incremental plan :232, :315); <code>frame_delta</code> , <code>frame_semantic_state</code> ; <code>frame_graph</code> = the verifier's plan <i>oracle</i> :668, :692-764.	compositor — passes, targets, blur / mask / opacity / clip <code>compositor_gpu.cljs</code>:1303	The family declaration table <code>frame_inputs.cljc</code>:28-69 with <code>:editor-pool-info</code> in the rect row; the ledger written to <code>js/globalThis</code> :242-248; two orphaned maintained twins — <code>maintain-frame-plan</code> <code>frame_graph.cljc</code>:677 and <code>maintain-effect-spans</code> <code>frame_effects.cljc</code>:250, test-only.	FLOOR · GOES the two orphans, the ledger · the table changes with the key	145
GPU core	<code>renderer/shape-text</code> — a real shaper: text ops + fonts → placed paint rects, placement backend-independent :2345-2353, :2277-2279.	shaders (≈18 % of the file), systems, frame compile / execute :3470-3677; <code>compositor</code> ; <code>buffer_pool</code> (slot / generation :120-141); <code>gpu_budget</code> ; <code>region_bindings</code>	The family registry :3297 + fence :3364 + buckets :3418; editor leftovers — <code>calculate-bracket-rects</code> :730, <code>hit-test</code> :750, <code>make-snapper</code> :2134, <code>create-editor-state</code> at 50 000 rects :2077, <code>draw-comparison-frame!</code> :2637; <code>gpu-mount</code> with its "DO NOT WIRE AS-IS" banner <code>buffer_pool.cljs</code>:448-457.	FLOOR · GOES the leftovers, <code>gpu-mount</code> · OPEN the key change	313

FAMILY	SAID → GEOMETRY (THE SHAPER)	PIPELINE	THE OLD-WORLD INSTANCE ON IT	WHERE IT SORTS	KB
the guard	CPU oracles — reference geometry for the pipelines: <code>region3d-lit-oracle verifier.cljs:4579-4648</code> , <code>point-in-curves? :317</code> , <code>route-boundary-distance :2482</code> .	67 golden files, headless puppeteer <code>test/render_engine/run_verifier.mjs</code>	Per-family fixture atoms — image 14 %, W4 frame 27 %, 3D 18 %, marks 6.5 %, arrows 9 % of the file; an estimated 60–70 % of it is scene construction (sampled, not counted).	<code>FLOOR</code> oracles + pipeline goldens · <code>GOES</code> the fixture atoms of whatever family goes; the W4 receipt rehomed	304
demos	—	—	<code>frame_runtime</code> : eight fixed trees at hard-coded coordinates + a PNG export chord <code>:61-163</code> , <code>:268-274</code> ; the 3D runtime's fixture; the <code>?live-atoms / ?region3d</code> readers nobody calls; <code>resources/public/faces/</code> .	<code>GOES</code>	12+

The cross-cutting instances

Statics — three kinds, not one

Go

- Providers set by mutation: `!effective-provider`, `!live-route-cache` `connector_route.cljc:18-21`; `!live-camera-provider`, `!live-effective-provider` `chrome_derive.cljc:23-24`
- Browser globals: `__softlandFrameLedger` `frame_inputs.cljc:242-248`, `window.region3d` `region3d_runtime.cljs:602-620`, `window.sceneContext` `scene_runtime.cljs:216`
- The runtimes' session atoms (19 across three files) — free, they go with the runtimes; the one store atom `scene_runtime.cljs:25`
- The 3D global `!pick-state` `region3d_gpu.cljs:706` — a cross-family mutable mirror

Stay below

- Device-owned resources: `!systems-by-device` `region3d_gpu.cljs:705`, `!compositors-by-device` `renderer.cljs:3397`
- System-local buffers and capacities (`buffer_pool` ; every `!buffer` / `!capacity`)
- Bounded derived caches (glyph maps `renderer.cljs:2147-2148`, mesh caches); budget accounting (`gpu_budget`); compositor leases
- Explicitly owned frame history where incremental rendering needs it (`frame_plan_view` state, `region_bindings`)

Only the *go* column is a build-stage item. "Thread every atom through as values" is a style choice with a per-call cost, not a ruling.

Gates — where "describe, never gate" applies

The law is about product descriptions, and it carves out the floor itself — *"Totality gates on the meta-material's OWN form (error cards) are the dull floor, not a violation"*

`docs/editable-material/DIRECTION.md:94`. So: **above**, an unknown facet still exists and gets an error / fallback projection — an untyped block renders, forever. **Below**, a road refuses a malformed packet (NaN, a bad index, an oversize mesh, a cyclic parent) and shows the fallback; the GPU is never handed invalid data.

Go — kind menus that are product vocabulary

- `legal-forms` `chrome_material.cljc:16`
- `legal-object-kinds` · `-primitive-` · `-light-` · `-camera-`
`region3d_material.cljc:18-22`
- `legal-route-policies` · `legal-heads`
`connector_material.cljc:14-17`

Stay — shape and totality checks

- `validate-region!` (finite vectors, indices, caps, acyclic parents)
`region3d_material.cljc:432-464`
- `contour / knot validation` `path_material.cljc:156`; entry shape
`scene_tape.cljc:653-688`
- the exhaustiveness fence `renderer.cljs:3364`

Twins — maintained view + oracle

The guard's pattern everywhere is an incremental view checked against a full recompute. An oracle is off the live path *by design* — it is not dead. Keep: `frame_graph` (the verifier's plan oracle, called six times from `verifier.cljs`), `frame_effects/derive-effect-spans`, `chrome_derive/full-recompute-oracle :231`, the scheduler's `decide` (the verifier's `verifier.cljs:4986`). Orphaned — only tests call them: `maintain-frame-plan` `frame_graph.cljc:677` · `frame_graph_test.clj:44-56`, `maintain-effect-spans` `frame_effects.cljc:250` · `frame_effects_test.clj:67-71`, and `gpu-mount` `buffer_pool.cljs:442-493`. Position: the three orphans go; the oracles stay with the guard.

Where this leaves the board

Cut — convergent, contract-ready at your word

Two independent reads, one list.

- the tree — `rect_tree` · `scene_store` · `scene_runtime`; `selection` with it
- the demos — `frame_runtime`, the 3D runtime's fixture, the two flag readers, `resources/public/faces/`; the W4 receipt rehomed into the verifier
- the marks vocabulary — five forms, `px` constants, `snap`, `chrome_derive`, `chrome_runtime`
- the small Blender — `region3d_runtime`, `region3d_pointer`, `orbit` / `gizmo` / `handle` picks + `edit ops` + `kind menus` in `_scene` / `_material`, `gizmo` + `grid` shaders, the global pick state — **regardless of the 3D answer**
- the three orphans; the ambient providers; the browser globals; the renderer's editor leftovers
- *position*, *vetoable*: the arrow family whole, with the lift list

Keep — floor candidates

- text shaping + layout + planes (+ the fonts loader)
- path authority + tessellation + the contour pipeline
- image preparation · `containers`
- scene ordering (the tape) · effects + plan + deltas
- compositor · pool · budget · region bindings
- the renderer's shaders, systems and frame execution
- road-level oracles + goldens
- the 3D kernel (mesh · hierarchy · BVH · ray · placement) — *if* a 3D facet is possible

Not yet contractable — build stage

Seen at the renderer's own grain, after the cuts.

- family key → facet key, with multi-facet entities (registry, declaration table, buckets, the per-family prepare / upload in `draw-frame!`)
- the render packet: road id + resource key; no live handle in `:paint/source`
- a form-free mark road (a neutral vertex input for `chrome_gpu`)
- `region3d_placement` 's input once the store is gone
- validator sorting — kind menus out, shape checks in
- the *go* statics · the legacy text road (needs a JVM shaped provider first) · the connector lift list, if the whole-family cut is vetoed

Only Sid

Can an entity wear a 3D mesh / material facet that needs depth-aware rendering and spatial queries?

Yes — the mesh road and the pure scene geometry stay. No — the 3D kernel goes too. The small Blender goes either way.

Position withheld on purpose: this one is yours, and dressing it in your past words was the thing you'd catch.

Sequence (both reviews, and I agree): rule the 3D facet → land this page → cut the instances → then look at the key change at the renderer's grain, with about 40 % less code to see through.

What is not sure

- Magnitudes — 60–70 % fixtures in the verifier, ~180 editor lines in `region3d_scene`, ≈18 % shader text in the renderer — are sampled estimates, not counts.
- The legacy text road: one JVM test asserts it; whether it is the *only* JVM road for layout is unchecked.
- `apply-variations!` `text_shaper.cljs:261` mutates HarfBuzz faces per call — variation axes may be order-dependent (a structure claim from reading, not a measured fault).
- The numerical checks Codex cites near `region3d_material.cljc:252` were not re-read here; `validate-region!` :432-464 was.

- Whether the connector's pure geometry is worth lifting or cheaper to rebuild is a build-stage call, not a shape fact.

Words

waist	your test (08-19): what has to exist as code vs what can be built on top, ECS or whatever style. The name is not yours — you asked why it was called that — the test is.
floor	the code under the waist; here, the render floor. Narrower than "must be code": reusable without Softland's meanings, stable geometric values, GPU custody.
facet · entity · recipe · system · binding · verb	your July words for ECS: a facet is one attached piece of data saying one true thing about an entity; a recipe is the list of facets a kind of thing wears; a system is code doing one job for everything wearing a facet.
family	the engine's own word for its ten kinds (<code>:render.family/*</code>) — the key this page says is the old world.
road · pipeline	session words. A pipeline is what exists on the GPU today (nine). A road is the target: one way to pixels keyed by one facet. Your phrases: "text in placement out"; "stroke shape → tessellator → triangles".
marks	your March word; the code says "chrome"; in the felt pass you said "the blue box", "bounding box".
the tape	session word (2026-08): the one ordered list of entries for a frame; paint walks it forward, pick walks it backward.
the small Blender	this page's phrase for the 3D window's editor — orbit, gizmo, lights menu, grid, edit ops, the draggable panel.
maintained view · oracle	the guard's pattern: an incremental computation and its from-scratch twin, checked against each other.

Evidence

Built from code, not from the prior maps: six Opus read-only gatherers by seam (GPU core · GPU families + verifier · frame machine · material families · workspace + the shaper yardstick · a docs-vs-code check), one gatherer indexing your own words (recall over every prompt you typed + `vision/LOG.md`), parent spot-checks on every decision-changing fact. Two

independent reviews were folded in — a sibling Claude session and Codex — on: the fence is not the class system · the move stated at its real size · quotes withdrawn where they carried a position · statics classified · gates re-scoped · twins made precise · the placement coupling · layout's purity · the 3D question restated · pick split into ray-hit and gizmo-hit.

The prior maps — `render-engine-map.md` , `engine-two-arrows.md` , `kept-code-map.md` — are evidence strata, not the current map: their relationship claims hold (eight of nine checked), their counts went stale within a day, and they rule while saying they don't ("leaks", CUSTODY as a delete). Your own words this page leans on: the waist test and "text in placement out" (08-19), "what should exist as code committed ... versus the ECS or small talk VM layer" (08-16), "i hate using the word type ... what we want is ecs style" (LOG 08-12), "a minimal VM-like floor beneath it" and the three strata (LOG 07-30), "why is box below the waist?? can it not be defined above the waist??" and the rect ruling (08-25).

The Engine at the Waist · docs/below-the-waist/engine-waist-map.md is the contestable twin of this page · sources: HEAD at ed84a6e, 2026-08-25 · positions are marked as positions; rulings are Sid's.